# The sharp $A_p$ bound for the Hardy–Littlewood maximal operator

An exposition of Lerner's proof of Buckley's theorem

Simon Gustafsson

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**Abstract**

In 1972, B. Muckenhoupt [5] proved that the Hardy–Littlewood maximal operator is bounded on the weighted space $L^p_w(\mathbb{R}^d)$, $1 < p < \infty$, if and only if $w$ satisfies the $A_p$ condition. In 1993, S. Buckley [1] showed that its operator norm is bounded by a constant times $[w]_{A_p}^{1/(p-1)}$, and that this exponent is best possible. In 2008, A. K. Lerner [4] gave an alternative proof of Buckley's bound. We present Lerner's proof in detail, together with Buckley's example showing that the exponent cannot be improved.

## 1 Introduction

**Definition 1** (Weights)**.** *A* weight *is a function $w \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ with $w \geq 0$. For a measurable set $E \subseteq \mathbb{R}^d$ we write $w(E) := \int_E w(x)\,dx$.*

**Definition 2** (Weighted $L^p$ spaces)**.** *Let $w$ be a weight and let $1 \leq p < \infty$. We define*

$$L^p_w(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} |f(x)|^p\, w(x)\,dx < \infty \right\},$$

*with norm $\|f\|_{L^p_w} := \left(\int_{\mathbb{R}^d} |f(x)|^p\, w(x)\,dx\right)^{1/p}$. Moreover, $L^\infty_w(\mathbb{R}^d)$ is the space of measurable functions $f$ with $\|f\|_{L^\infty_w} := \inf\{t \geq 0 : w(\{|f| > t\}) = 0\} < \infty$.*

The central object of this report is the Hardy–Littlewood maximal operator $M$. Our proofs also use its analogue with respect to the measure $w(x)\,dx$.

**Definition 3** (Hardy–Littlewood maximal operator)**.** *Let $w$ be a weight with $w(Q) > 0$ for every cube $Q$. For a measurable function $f$ and $x \in \mathbb{R}^d$,*

$$M_w f(x) := \sup_{Q \ni x} \frac{1}{w(Q)} \int_Q |f(y)|\, w(y)\,dy,$$

*where the supremum is taken over all closed cubes $Q \subset \mathbb{R}^d$ with sides parallel to the axes containing $x$. All cubes in this report are of this kind. If the supremum is taken only over the cubes $Q(x,r) := \{y : |y - x|_\infty \leq r/2\}$, $r > 0$, where $|y|_\infty := \max_{1 \leq i \leq d} |y_i|$, we write $M^c_w$. If $w \equiv 1$ we drop the subscript and write $M$ and $M^c$.*

It is well known that $M$ is of weak type $(1,1)$ and strong type $(p,p)$ for $1 < p \leq \infty$ [3, Theorem 1.1]. For $1 < p < \infty$ and a weight $w$ we say that $M$ is bounded on $L^p_w(\mathbb{R}^d)$ if there is a constant $C$ such that

$$\|Mf\|_{L^p_w} = \left( \int_{\mathbb{R}^d} (Mf)^p \, w \, dx \right)^{1/p} \leq C \left( \int_{\mathbb{R}^d} |f|^p \, w \, dx \right)^{1/p} = C \, \|f\|_{L^p_w} \tag{1}$$

for all $f \in L^p_w(\mathbb{R}^d)$.

**Remark 1** (The zero weight)**.** *If $w = 0$ almost everywhere, then $\|f\|_{L^p_w} = 0$ for every $f$, so $L^p_w(\mathbb{R}^d)$ is trivial and $M$ is trivially bounded.*

**Remark 2** (The endpoint $p = \infty$)**.** *For $p = \infty$, the operator $M$ is bounded on $L^\infty_w(\mathbb{R}^d)$ if and only if $w > 0$ almost everywhere or $w = 0$ almost everywhere. Muckenhoupt [5, Theorem 3] proves this on an interval, and the same argument works in $\mathbb{R}^d$. This is why we only consider $1 < p < \infty$.*

Suppose that (1) holds and let $Q$ be a cube. Since $Mf(x)$ is at least the average of $|f|$ over $Q$ for $x \in Q$, writing out the norms in (1) by Definition 2 gives

$$\left( \frac{1}{|Q|} \int_Q |f| \, dx \right)^p w(Q) \leq \int_Q (Mf)^p \, w \, dx \leq C^p \int_{\mathbb{R}^d} |f|^p \, w \, dx$$

for every $f \in L^p_w(\mathbb{R}^d)$. Each choice of $f$ thus gives a condition on $w$, and the choice $f = w^{-\frac{1}{p-1}} \chi_Q$ makes this condition as strong as possible. The following proposition makes this precise, and we prove it in Appendix A.

**Proposition 1.** *Let $1 < p < \infty$ and let $w$ be a weight that is not almost everywhere zero. If (1) holds with constant $C$, then for every cube $Q$ we have $\int_Q w^{-\frac{1}{p-1}} \, dx < \infty$ and*

$$\left( \frac{1}{|Q|} \int_Q w \, dx \right) \left( \frac{1}{|Q|} \int_Q w^{-\frac{1}{p-1}} \, dx \right)^{p-1} \leq C^p. \tag{2}$$

*In particular, $w > 0$ almost everywhere.*

This motivates the following definition [5, (7.10)].

**Definition 4** (Muckenhoupt $A_p$ weights)**.** *Let $1 < p < \infty$ and let $w$ be a weight. The $A_p$* characteristic *of $w$ is*

$$[w]_{A_p} := \sup_Q \left( \frac{1}{|Q|} \int_Q w(x) \, dx \right) \left( \frac{1}{|Q|} \int_Q w(x)^{-\frac{1}{p-1}} \, dx \right)^{p-1}, \tag{3}$$

*where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$. We say that $w \in A_p$ if $[w]_{A_p} < \infty$.*

**Definition 5** (Dual weight)**.** *Let $w$ be a weight and let $1 < p < \infty$. The* dual weight *of $w$ is $\sigma := w^{-\frac{1}{p-1}}$, with $\sigma = \infty$ where $w = 0$ [1, Section 1]. For measurable sets $E$ we write $\sigma(E) := \int_E \sigma \, dx$.*

With this notation (3) takes the more compact form

$$[w]_{A_p} = \sup_Q \frac{w(Q)}{|Q|} \left( \frac{\sigma(Q)}{|Q|} \right)^{p-1}.$$

Taking the supremum over all cubes in Proposition 1 gives the following.

**Corollary 1.** *Let $1 < p < \infty$ and let $w$ be a weight that is not almost everywhere zero. If (1) holds with constant $C$, then $w \in A_p$ and*

$$[w]_{A_p} \leq C^p. \tag{4}$$

Muckenhoupt proved that the condition $w \in A_p$ is also sufficient.

**Theorem A** (Muckenhoupt [5, Theorem 9])**.** *Let $1 < p < \infty$ and let $w$ be a weight that is not almost everywhere zero. Then $M$ is bounded on $L^p_w(\mathbb{R}^d)$ if and only if $w \in A_p$.*

Theorem A shows that the operator norm of $M$ on $L^p_w(\mathbb{R}^d)$, that is, the smallest constant $C$ in (1), is finite precisely when $w \in A_p$. By Corollary 1, this norm is bounded from below by $[w]_{A_p}^{1/p}$. The natural question is therefore whether it is also bounded from above by a power of $[w]_{A_p}$, and which power is the smallest possible. This was answered by Buckley [1, Theorem 2.5].

**Theorem 1** (Buckley)**.** *Let $1 < p < \infty$. There is a constant $c_{p,d}$, depending only on $p$ and $d$, such that for every $w \in A_p$ and every $f \in L^p_w(\mathbb{R}^d)$,*

$$\|Mf\|_{L^p_w} \leq c_{p,d}\, [w]_{A_p}^{\frac{1}{p-1}}\, \|f\|_{L^p_w}.$$

**Proposition 2** (Buckley)**.** *Let $1 < p < \infty$ and $\alpha < \frac{1}{p-1}$. Then there is no constant $c$ such that $\|Mf\|_{L^p_w} \leq c\, [w]_{A_p}^{\alpha}\, \|f\|_{L^p_w}$ for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$.*

Buckley's proof of Theorem 1 [1, proof of Theorem 2.5], like Muckenhoupt's proof of Theorem A [5, Lemma 5 and Theorem 2], relies on a deep property that every $A_p$ weight belongs to $A_{p-\varepsilon}$ for some $\varepsilon > 0$ [3, Theorem 4.5], together with interpolation. Lerner [4, Theorem C] found a short and elementary proof of Theorem 1. We present his argument in detail and complete it with a proof of Proposition 2, based on Buckley's example in $\mathbb{R}$ [1, proof of Theorem 2.5], which we extend to $\mathbb{R}^d$.

**Remark 3** (An alternative proof of Theorem A)**.** *If $M$ is bounded on $L^p_w(\mathbb{R}^d)$, then $w \in A_p$ by Corollary 1, which follows from Proposition 1. Conversely, if $w \in A_p$, then $M$ is bounded on $L^p_w(\mathbb{R}^d)$ by Theorem 1. Combined, these results imply Theorem A.*

# 2 Proofs

## 2.1 Preliminaries

**Lemma 1.** *Let $Q$ be a cube with side length $\ell(Q)$ and let $y \in Q$. Then $Q \subseteq Q(y, 2\ell(Q)) \subseteq 3Q$, where $3Q$ denotes the cube with the same centre as $Q$ and side length $3\ell(Q)$.*

*Proof.* Every $z \in Q$ satisfies $|z - y|_\infty \leq \ell(Q)$, since both $z$ and $y$ lie in $Q$. Hence $Q \subseteq Q(y, 2\ell(Q))$. If $c$ is the centre of $Q$, then $|y - c|_\infty \leq \frac{1}{2}\ell(Q)$, so every $z \in Q(y, 2\ell(Q))$ satisfies $|z - c|_\infty \leq |z - y|_\infty + |y - c|_\infty \leq \frac{3}{2}\ell(Q)$, that is, $z \in 3Q$. □

**Lemma 2.** *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. Then $Mf(x) \leq 2^d M^c f(x)$ for every $x \in \mathbb{R}^d$.*

*Proof.* Let $x \in \mathbb{R}^d$. For a cube $Q$ containing $x$, Lemma 1 gives $Q \subseteq Q(x, 2\ell(Q))$, and since $|Q(x, 2\ell(Q))| = 2^d|Q|$ we obtain

$$\begin{aligned} Mf(x) &= \sup_{Q \ni x} \frac{1}{|Q|} \int_Q |f(y)|\, dy \\ &\leq \sup_{Q \ni x} \frac{2^d}{|Q(x, 2\ell(Q))|} \int_{Q(x,2\ell(Q))} |f(y)|\, dy \\ &\leq 2^d \sup_{r>0} \frac{1}{|Q(x,r)|} \int_{Q(x,r)} |f(y)|\, dy \\ &= 2^d M^c f(x). \end{aligned}$$

□

**Lemma 3.** *Let $1 < p < \infty$ and let $w \in A_p$ be a weight that is not almost everywhere zero. Then $w > 0$ almost everywhere and $\sigma$, as in Definition 5, is a weight. Moreover $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube $Q$.*

*Proof.* Let $Q$ be a cube with $w(Q) > 0$. By the definition of $[w]_{A_p}$,

$$\left(\frac{\sigma(Q)}{|Q|}\right)^{p-1} \leq [w]_{A_p} \frac{|Q|}{w(Q)} < \infty, \tag{5}$$

so $\sigma(Q) < \infty$ on any cube $Q$ where $w(Q) > 0$.

Assume, for contradiction, that $w = 0$ on a set $E$ with $|E| > 0$. Since $w$ is not almost everywhere zero, $w > 0$ on a set $F$ with $|F| > 0$, so we can choose a cube $Q$ with $|Q \cap E| > 0$ and $|Q \cap F| > 0$. Then $w(Q) \geq w(Q \cap F) > 0$, so $\sigma(Q) < \infty$ by (5). But $\sigma = \infty$ on $Q \cap E$ by Definition 5, so $\sigma(Q) = \infty$, a contradiction. Therefore $w > 0$ almost everywhere. In particular $w(Q) > 0$, and hence $\sigma(Q) < \infty$ by (5), for every cube $Q$.

Since $w(Q) < \infty$ for every cube $Q$, we have $w < \infty$ almost everywhere, so $\sigma > 0$ almost everywhere and $\sigma(Q) > 0$ for every cube $Q$. As every compact set lies in a cube, $\sigma$ is locally integrable and hence a weight. □

**Remark 4** (The Vitali covering lemma)**.** *In the proof of the weak type $(1,1)$ estimate for $M$ [3, Theorem 1.1], a compact set $K \subseteq \{Mf > \lambda\}$ is covered by finitely many cubes $Q$ with $\frac{1}{|Q|}\int_Q |f|\, dx > \lambda$. The Vitali covering lemma [3, Lemma A.4], which holds for balls as well as cubes, selects disjoint cubes $Q_1, \ldots, Q_N$ among them with $K \subseteq \bigcup_j 3Q_j$. Since $|3Q_j| = 3^d|Q_j|$, this gives*

$$|K| \leq \sum_{j=1}^N |3Q_j| = 3^d \sum_{j=1}^N |Q_j| \leq \frac{3^d}{\lambda} \int_{\mathbb{R}^d} |f|\, dx.$$

*For the measure $v(x)\, dx$, the equality $|3Q_j| = 3^d|Q_j|$ would have to be replaced by an inequality $v(3Q_j) \leq C\, v(Q_j)$, and no constant $C$ works for all weights $v$. For example, if $v = \varepsilon$ on $Q_j$ and $v = 1$ on the rest of $3Q_j$, then $v(3Q_j)/v(Q_j) \geq (3^d - 1)/\varepsilon$, which is arbitrarily large. The Besicovitch covering lemma covers $K$ by the selected cubes themselves, without enlarging them, and therefore gives a constant independent of $v$.*

**Lemma 4** (Besicovitch covering lemma [2, Theorem 1.1])**.** *Let $E \subset \mathbb{R}^d$ be a bounded set, and suppose that for each $x \in E$ a cube $Q(x, r_x)$ centred at $x$ is given. Then there are points $x_1, x_2, \ldots \in E$ such that the cubes $Q(x_j, r_{x_j})$ cover $E$ and every point of $\mathbb{R}^d$ lies in at most $N_d$ of them, where $N_d$ depends only on $d$.*

*Proof.* We may assume that $E$ is nonempty, and we let $D := \sup_{x,y\in E} |x-y|_\infty < \infty$. If $r_x \geq 2D$ for some $x \in E$, then $E \subseteq Q(x, r_x)$ and this single cube suffices. Hence we may assume that $r_x < 2D$ for every $x \in E$.

We choose points $x_1, x_2, \ldots \in E$ inductively. Let $m \geq 0$ and suppose that $x_1, \ldots, x_m$ have been chosen. If $E \setminus \bigcup_{j\leq m} Q(x_j, r_{x_j})$ is empty, we stop. Otherwise we let $s_m := \sup\{r_x : x \in E \setminus \bigcup_{j\leq m} Q(x_j, r_{x_j})\}$ and choose $x_{m+1} \in E \setminus \bigcup_{j\leq m} Q(x_j, r_{x_j})$ with $r_{x_{m+1}} > s_m/2$. If $j < i$, then $x_i \notin Q(x_j, r_{x_j})$, and $x_i$ was also a possible choice when $x_j$ was chosen, so

$$|x_i - x_j|_\infty > \frac{r_{x_j}}{2} \qquad \text{and} \qquad r_{x_i} \leq s_{j-1} < 2r_{x_j}. \tag{6}$$

If the selection stops, the cubes $Q(x_j, r_{x_j})$ cover $E$ by construction. Assume that it does not stop, and suppose, for contradiction, that some $x \in E$ is not covered. Then $x$ is a possible choice at every step, so $r_x \leq s_{j-1} < 2r_{x_j}$ for every $j$. If $j < i$, then by (6)

$$\frac{r_{x_i}}{8} + \frac{r_{x_j}}{8} < \frac{3r_{x_j}}{8} < |x_i - x_j|_\infty,$$

so the cubes $Q(x_j, r_{x_j}/4)$ are pairwise disjoint. Since $x_j \in E$ and $r_{x_j} < 2D$, they are all contained in $Q(x_1, 4D)$. Comparing volumes, for every $N$,

$$N\left(\frac{r_x}{8}\right)^d < \sum_{j=1}^{N} \left(\frac{r_{x_j}}{4}\right)^d \leq (4D)^d,$$

which is false for large $N$. Hence every $x \in E$ is covered.

It remains to bound the overlap. Fix $z \in \mathbb{R}^d$. The space $\mathbb{R}^d$ is the union of the $2^d$ closed orthants with vertex $z$, that is, the sets $\{y : \varepsilon_i(y_i - z_i) \geq 0 \text{ for } i = 1, \ldots, d\}$ with $\varepsilon \in \{-1, 1\}^d$. If $y$ and $y'$ lie in the same orthant, then $y_i - z_i$ and $y'_i - z_i$ have the same sign for every $i$, and hence

$$|y - y'|_\infty \leq \max\{|y - z|_\infty,\ |y' - z|_\infty\}. \tag{7}$$

Fix one orthant and let $J$ be the set of indices $i$ such that $x_i$ lies in this orthant and $z \in Q(x_i, r_{x_i})$. Suppose that $J$ is nonempty and let $j$ be its smallest element. Let $i \in J$ with $i > j$. Since $|x_j - z|_\infty \leq r_{x_j}/2$, (6) and (7) give

$$\frac{r_{x_j}}{2} < |x_i - x_j|_\infty \leq \max\{|x_i - z|_\infty,\ |x_j - z|_\infty\} = |x_i - z|_\infty \leq \frac{r_{x_i}}{2},$$

so $r_{x_i} > r_{x_j}$, and by (6) also $|x_i - z|_\infty \leq r_{x_i}/2 < r_{x_j}$. Moreover, if $i, i' \in J$ with $i > i' > j$, then (6) gives $|x_i - x_{i'}|_\infty > r_{x_{i'}}/2 > r_{x_j}/2$. Hence the cubes $Q(x_i, r_{x_j}/2)$ with $i \in J$ and $i > j$ are pairwise disjoint, and they are contained in $Q(z, 5r_{x_j}/2)$. Comparing volumes, there are at most $5^d$ such indices $i$, so $J$ has at most $1 + 5^d$ elements. Summing over the $2^d$ orthants, $z$ lies in at most $2^d(1+5^d)$ of the cubes $Q(x_j, r_{x_j})$, and the lemma holds with $N_d := 2^d(1 + 5^d)$. □

**Lemma 5.** *Let $v$ be a weight with $v(Q) > 0$ for every cube $Q$, and let $f$ be a measurable function such that $|f|\, v$ is locally integrable. Then $M_v^c f$ is measurable, and*

$$v\big(\{x \in \mathbb{R}^d : M_v^c f(x) > \lambda\}\big) \leq \frac{N_d}{\lambda} \|f\|_{L^1_v}$$

*for every $\lambda > 0$, where $N_d$ is the constant in Lemma 4.*

*Proof.* Fix $\lambda > 0$, and call a cube $Q(x,r)$ *good* if

$$\int_{Q(x,r)} |f(y)|\, v(y)\, dy > \lambda\, v(Q(x,r)). \tag{8}$$

By the definition of $M_v^c$, we have $M_v^c f(x) > \lambda$ if and only if $Q(x,r)$ is good for some $r > 0$. For fixed $r > 0$, both sides of (8) are continuous functions of $x$. Indeed, if $x' \to x$, then $\chi_{Q(x',r)} \to \chi_{Q(x,r)}$ outside the boundary of $Q(x,r)$, which has measure zero, and $\chi_{Q(x',r)} \le \chi_{Q(x,r+2)}$ whenever $|x' - x| \le 1$, so the claim follows by dominated convergence since $|f|\, v$ and $v$ are integrable on $Q(x, r+2)$. Hence the set $U_r$ of all $x$ for which $Q(x,r)$ is good is open, and so is $\{M_v^c f > \lambda\} = \bigcup_{r>0} U_r$. Since $\lambda > 0$ was arbitrary and $M_v^c f \ge 0$, the function $M_v^c f$ is measurable.

For every positive integer $k$ let $E_k := \{x : |x| < k \text{ and } M_v^c f(x) > \lambda\}$. For each $x \in E_k$ we choose $r_x > 0$ such that $Q(x, r_x)$ is good. Since $E_k$ is bounded, Lemma 4 gives points $x_1, x_2, \ldots \in E_k$ such that the cubes $Q(x_j, r_{x_j})$ cover $E_k$ and satisfy $\sum_j \chi_{Q(x_j, r_{x_j})} \le N_d$. Using (8) for each $Q(x_j, r_{x_j})$ and monotone convergence to interchange sum and integral, we obtain

$$\begin{aligned} v(E_k) &\le \sum_j v\big(Q(x_j, r_{x_j})\big) \le \frac{1}{\lambda} \sum_j \int_{Q(x_j, r_{x_j})} |f|\, v\, dy \\ &= \frac{1}{\lambda} \int_{\mathbb{R}^d} \Big( \sum_j \chi_{Q(x_j, r_{x_j})} \Big) |f|\, v\, dy \le \frac{N_d}{\lambda}\, \|f\|_{L_v^1}. \end{aligned}$$

Since $E_1 \subseteq E_2 \subseteq \cdots$ and $\bigcup_k E_k = \{M_v^c f > \lambda\}$, continuity of the measure $v(x)\, dx$ from below gives the claim. □

**Lemma 6.** *Let $v$ be a weight with $v(Q) > 0$ for every cube $Q$, and let $1 < q < \infty$. Then there is a constant $C_{q,d}$, depending only on $q$ and $d$, such that*

$$\|M_v^c f\|_{L_v^q} \le C_{q,d}\, \|f\|_{L_v^q}$$

*for all $f \in L_v^q(\mathbb{R}^d)$.*

*Proof.* If $f = f_1 + f_2$ with $f_1 \in L_v^1(\mathbb{R}^d)$ and $f_2 \in L_v^\infty(\mathbb{R}^d)$, then $\int_Q |f|\, v\, dy \le \|f_1\|_{L_v^1} + \|f_2\|_{L_v^\infty}\, v(Q) < \infty$ for every cube $Q$. Hence $M_v^c$ is defined on $L_v^1(\mathbb{R}^d) + L_v^\infty(\mathbb{R}^d)$, and $M_v^c f$ is measurable by Lemma 5. Since $|f + h| \le |f| + |h|$, the operator $M_v^c$ is sublinear.

By Lemma 5, $v(\{M_v^c f > \lambda\}) \le \frac{N_d}{\lambda} \|f\|_{L_v^1}$ for every $f \in L_v^1(\mathbb{R}^d)$ and $\lambda > 0$. Moreover, if $f \in L_v^\infty(\mathbb{R}^d)$, then $|f| \le \|f\|_{L_v^\infty}$ outside a set of $v$-measure zero, so every average in the definition of $M_v^c f$ is at most $\|f\|_{L_v^\infty}$, and $\|M_v^c f\|_{L_v^\infty} \le \|f\|_{L_v^\infty}$. The Marcinkiewicz interpolation theorem [3, Theorem A.6], applied on the measure space $(\mathbb{R}^d, v(x)\, dx)$ with exponents 1 and $\infty$, now gives the claim with a constant depending only on $q$, $N_d$ and 1, and hence only on $q$ and $d$. □

## 2.2 Proof of Theorem 1

*Proof of Theorem 1.* Let $1 < p < \infty$, let $w \in A_p$ and let $f \in L_w^p(\mathbb{R}^d)$. If $w = 0$ almost everywhere, both sides of the inequality vanish, so we may assume that $w$ is not almost everywhere zero. By Lemma 3, $w > 0$ almost everywhere and $\sigma$ is a weight with $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube $Q$. In particular, the operators $M_w^c$ and $M_\sigma^c$ are defined.

We first prove a pointwise estimate for $Mf$. Fix $x \in \mathbb{R}^d$ and $r > 0$, let $Q := Q(x,r)$, and let $3Q$ be as in Lemma 1. Multiplying and dividing by $w(Q)\,\sigma(3Q)^{p-1}$, we obtain

$$\begin{aligned}\left(\frac{1}{|Q|}\int_Q |f(y)|\,dy\right)^{p-1} &= \frac{1}{|Q|^{p-1}}\left(\int_Q |f(y)|\,dy\right)^{p-1}\\ &= \frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p}\cdot\frac{|Q|}{w(Q)}\cdot\frac{1}{\sigma(3Q)^{p-1}}\left(\int_Q |f(y)|\,dy\right)^{p-1}\\ &= \frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p}\cdot\frac{|Q|}{w(Q)}\left(\frac{1}{\sigma(3Q)}\int_Q |f(y)|\,dy\right)^{p-1}.\end{aligned} \tag{9}$$

Since $w(Q) \le w(3Q)$ and $|3Q| = 3^d|Q|$, the first factor on the right-hand side satisfies

$$\frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p} \le 3^{dp}\left(\frac{1}{|3Q|}\int_{3Q} w(y)\,dy\right)\left(\frac{1}{|3Q|}\int_{3Q}\sigma(y)\,dy\right)^{p-1} \le 3^{dp}\,[w]_{A_p}.$$

For the second factor, let $y \in Q$. By Lemma 1, $Q \subseteq Q(y, 2\ell(Q)) \subseteq 3Q$, so

$$\begin{aligned}\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz &= \frac{1}{\sigma(3Q)}\int_Q |f(z)|\,\sigma(z)^{-1}\,\sigma(z)\,dz\\ &\le \frac{1}{\sigma(Q(y,2\ell(Q)))}\int_{Q(y,2\ell(Q))} |f(z)|\,\sigma(z)^{-1}\,\sigma(z)\,dz \le M_\sigma^c\big(|f|\sigma^{-1}\big)(y).\end{aligned}$$

This holds for every $y \in Q$, and therefore

$$\begin{aligned}\frac{|Q|}{w(Q)}\left(\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz\right)^{p-1} &= \frac{1}{w(Q)}\int_Q\left(\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz\right)^{p-1} dy\\ &\le \frac{1}{w(Q)}\int_Q \Big(M_\sigma^c\big(|f|\sigma^{-1}\big)(y)\Big)^{p-1} w(y)^{-1}\,w(y)\,dy\\ &\le M_w^c\Big(\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1}w^{-1}\Big)(x),\end{aligned}$$

where the last inequality holds because $Q = Q(x,r)$. Inserting both bounds into (9), taking the supremum over $r > 0$ and applying Lemma 2, we arrive at

$$Mf(x) \le 2^d\,3^{\frac{dp}{p-1}}\,[w]_{A_p}^{\frac{1}{p-1}}\Big(M_w^c\Big(\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1}w^{-1}\Big)(x)\Big)^{\frac{1}{p-1}}. \tag{10}$$

We now take norms. Since $1 - \frac{p}{p-1} = -\frac{1}{p-1}$, we have $w^{1-\frac{p}{p-1}} = \sigma$, and similarly $\sigma^{1-p} = w$. The second identity gives

$$\big\||f|\sigma^{-1}\big\|_{L^p_\sigma}^p = \int_{\mathbb{R}^d} |f(x)|^p\,\sigma(x)^{-p}\,\sigma(x)\,dx = \int_{\mathbb{R}^d} |f(x)|^p\,w(x)\,dx = \|f\|_{L^p_w}^p,$$

so $|f|\sigma^{-1} \in L^p_\sigma(\mathbb{R}^d)$. Using the first identity and Lemma 6 with $v = \sigma$ and $q = p$, we obtain

$$\begin{aligned}\int_{\mathbb{R}^d}\Big(\big(M_\sigma^c(|f|\sigma^{-1})(x)\big)^{p-1}w(x)^{-1}\Big)^{\frac{p}{p-1}}w(x)\,dx &= \int_{\mathbb{R}^d}\big(M_\sigma^c(|f|\sigma^{-1})(x)\big)^p\,\sigma(x)\,dx\\ &\le C_{p,d}^p\,\|f\|_{L^p_w}^p,\end{aligned}$$

so the function $\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1} w^{-1}$ lies in $L_w^{p/(p-1)}(\mathbb{R}^d)$. Finally, we raise (10) to the power $p$, integrate against $w$, and apply Lemma 6 with $v = w$ and $q = \frac{p}{p-1}$. Together with the previous estimate this gives

$$\|Mf\|_{L_w^p}^p \le \Big(2^d\, 3^{\frac{dp}{p-1}}\Big)^p\, [w]_{A_p}^{\frac{p}{p-1}}\, C_{\frac{p}{p-1},d}^{\frac{p}{p-1}}\, C_{p,d}^p\, \|f\|_{L_w^p}^p.$$

Taking $p$-th roots, we conclude that

$$\|Mf\|_{L_w^p} \le c_{p,d}\, [w]_{A_p}^{\frac{1}{p-1}}\, \|f\|_{L_w^p}, \qquad c_{p,d} := 2^d\, 3^{\frac{dp}{p-1}}\, C_{\frac{p}{p-1},d}^{\frac{1}{p-1}}\, C_{p,d},$$

and $c_{p,d}$ depends only on $p$ and $d$. □

## 2.3 Sharpness of the exponent

*Proof of Proposition 2.* Let $1 < p < \infty$ and $\alpha < \frac{1}{p-1}$, and suppose, for contradiction, that there is a constant $c$ such that

$$\|Mf\|_{L_w^p} \le c\, [w]_{A_p}^{\alpha}\, \|f\|_{L_w^p} \tag{11}$$

for all $w \in A_p$ and all $f \in L_w^p(\mathbb{R}^d)$. For every weight $w$ that is not almost everywhere zero and every cube $Q$, Hölder's inequality with exponents $p$ and $\frac{p}{p-1}$ gives

$$1 = \frac{1}{|Q|}\int_Q w^{\frac{1}{p}}\, w^{-\frac{1}{p}}\, dy \le \left(\frac{1}{|Q|}\int_Q w\, dy\right)^{\frac{1}{p}} \left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\, dy\right)^{\frac{p-1}{p}},$$

so $[w]_{A_p} \ge 1$. Hence (11) remains true if $\alpha$ is replaced by $\max\{\alpha, 0\}$, and we may assume that $0 \le \alpha < \frac{1}{p-1}$.

For $0 < \delta < 1$ we consider the weight $w_\delta(x) := |x|^{(d-\delta)(p-1)}$, for which $w_\delta^{-\frac{1}{p-1}}(x) = |x|^{\delta-d}$. By polar coordinates,

$$\int_{\{|y|<r\}} |y|^{\delta-d}\, dy = \omega_d\, \frac{r^\delta}{\delta} \qquad \text{for every } r > 0, \tag{12}$$

where $\omega_d$ denotes the surface measure of the unit sphere in $\mathbb{R}^d$.

We first show that $w_\delta \in A_p$ with

$$[w_\delta]_{A_p} \le C\, \delta^{1-p}, \tag{13}$$

where $C$ depends only on $p$ and $d$. The function $w_\delta$ is continuous and positive outside the origin, so it is a weight that is not almost everywhere zero. Write $a := (d-\delta)(p-1) \ge 0$, and let $Q := Q(c, \ell)$ be a cube. Every $y \in Q$ satisfies $|y - c| \le \frac{\sqrt{d}}{2}\, \ell$.

Suppose first that $|c| \le \sqrt{d}\, \ell$. Then every $y \in Q$ satisfies $|y| \le R\ell$ with $R := \frac{3\sqrt{d}}{2}$. Since $a \ge 0$, this gives $w_\delta \le (R\ell)^a$ on $Q$, and by (12),

$$\frac{1}{|Q|}\int_Q |y|^{\delta-d}\, dy \le \frac{1}{\ell^d}\int_{\{|y|<R\ell\}} |y|^{\delta-d}\, dy = \frac{\omega_d\, R^\delta\, \ell^{\delta-d}}{\delta}.$$

Since $a + (\delta - d)(p-1) = 0$, we obtain

$$\left(\frac{1}{|Q|}\int_Q w_\delta\, dy\right)\left(\frac{1}{|Q|}\int_Q |y|^{\delta-d}\, dy\right)^{p-1} \le (R\ell)^a \left(\frac{\omega_d\, R^\delta\, \ell^{\delta-d}}{\delta}\right)^{p-1} = R^{d(p-1)}\, \omega_d^{p-1}\, \delta^{1-p}.$$

Suppose instead that $|c| > \sqrt{d}\,\ell$. Then $|y-c| < \frac{1}{2}|c|$, so $\frac{1}{2}|c| \le |y| \le \frac{3}{2}|c|$ for every $y \in Q$. Hence $w_\delta \le \left(\frac{3}{2}|c|\right)^a$ and $|y|^{\delta-d} \le \left(\frac{1}{2}|c|\right)^{\delta-d}$ on $Q$, and

$$\left(\frac{1}{|Q|}\int_Q w_\delta\,dy\right)\left(\frac{1}{|Q|}\int_Q |y|^{\delta-d}\,dy\right)^{p-1} \le \left(\tfrac{3}{2}\right)^a 2^{(d-\delta)(p-1)} = 3^a \le 3^{d(p-1)} \le 3^{d(p-1)}\,\delta^{1-p}.$$

Taking the supremum over all cubes proves (13) with $C := \max\{R^{d(p-1)}\omega_d^{p-1},\, 3^{d(p-1)}\}$.

Next we construct a function for which $M$ is large in $L^p_{w_\delta}$. Let $f_\delta(x) := |x|^{\delta-d}\chi_{Q(0,2)}(x)$. Then $f_\delta \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, and since $Q(0,2) \subseteq \{|y| < 2\sqrt{d}\}$, (12) gives

$$\|f_\delta\|^p_{L^p_{w_\delta}} = \int_{Q(0,2)} |y|^{(\delta-d)p}\,|y|^{(d-\delta)(p-1)}\,dy = \int_{Q(0,2)} |y|^{\delta-d}\,dy \le \omega_d\,\frac{(2\sqrt{d})^\delta}{\delta} < \infty.$$

Moreover $\|f_\delta\|_{L^p_{w_\delta}} > 0$, so $f_\delta \in L^p_{w_\delta}(\mathbb{R}^d)$ is not the zero element. Let $x \in Q(0,2)$ with $x \neq 0$, and put $r := \max_{1\le i\le d}|x_i|$, so that $0 < r \le |x|$. The cube $Q(0,2r)$ contains $x$, is contained in $Q(0,2)$, and contains the ball $\{|y| < r\}$. By the definition of $M$ and (12),

$$Mf_\delta(x) \ge \frac{1}{(2r)^d}\int_{Q(0,2r)} |y|^{\delta-d}\,dy \ge \frac{1}{(2r)^d}\int_{\{|y|<r\}} |y|^{\delta-d}\,dy = \frac{\omega_d}{2^d\delta}\,r^{\delta-d} \ge \frac{\omega_d}{2^d\delta}\,|x|^{\delta-d},$$

where the last inequality holds because $\delta - d < 0$ and $r \le |x|$. Thus $Mf_\delta \ge \frac{\omega_d}{2^d\delta} f_\delta$ almost everywhere on $Q(0,2)$, and this also holds outside $Q(0,2)$, where $f_\delta = 0$. Consequently

$$\|Mf_\delta\|_{L^p_{w_\delta}} \ge \frac{\omega_d}{2^d\delta}\,\|f_\delta\|_{L^p_{w_\delta}}. \tag{14}$$

Finally we combine these estimates. Applying (11) to $w_\delta$ and $f_\delta$, and using (14), (13) and $\alpha \ge 0$, we obtain

$$\frac{\omega_d}{2^d\delta}\,\|f_\delta\|_{L^p_{w_\delta}} \le \|Mf_\delta\|_{L^p_{w_\delta}} \le c\,[w_\delta]^\alpha_{A_p}\,\|f_\delta\|_{L^p_{w_\delta}} \le c\,C^\alpha\,\delta^{\alpha(1-p)}\,\|f_\delta\|_{L^p_{w_\delta}}.$$

Dividing by $\|f_\delta\|_{L^p_{w_\delta}} \in (0,\infty)$ and multiplying by $\delta$ gives

$$\frac{\omega_d}{2^d} \le c\,C^\alpha\,\delta^{1-\alpha(p-1)} \qquad \text{for every } 0 < \delta < 1.$$

Since $\alpha < \frac{1}{p-1}$, the exponent $1-\alpha(p-1)$ is positive, so the right-hand side tends to 0 as $\delta \to 0$, while the left-hand side is a positive constant independent of $\delta$. This is a contradiction, and the proof is complete. □

# A Proof of Proposition 1

We follow the proof of [3, Theorem 1.7].

*Proof.* We first show that $w > 0$ almost everywhere. Fix a cube $Q$ and a measurable set $S \subseteq Q$. For $x \in Q$ the cube $Q$ is one of the cubes in the supremum defining $M\chi_S(x)$, so $M\chi_S(x) \geq |S|/|Q|$. Hence (1) applied to $f = \chi_S$ gives

$$\left(\frac{|S|}{|Q|}\right)^p w(Q) \leq C^p\, w(S).$$

Since $w$ is not almost everywhere zero, every bounded set $S$ is contained in a cube $Q$ with $w(Q) > 0$, so $w(S) = 0$ implies $|S| = 0$. Applying this to the sets $\{w = 0\} \cap \{|x| < k\}$, $k = 1, 2, \ldots$, shows that $w > 0$ almost everywhere. In particular $w(Q) > 0$ for every cube $Q$.

Now fix a cube $Q$ and $\varepsilon > 0$, and let $f := (w+\varepsilon)^{-\frac{1}{p-1}}\,\chi_Q$. Since $w \leq w + \varepsilon$,

$$(w+\varepsilon)^{-\frac{p}{p-1}}\, w \leq (w+\varepsilon)^{-\frac{p}{p-1}}\,(w+\varepsilon) = (w+\varepsilon)^{-\frac{1}{p-1}} \leq \varepsilon^{-\frac{1}{p-1}},$$

so $f \in L^p_w(\mathbb{R}^d)$. Moreover, for $x \in Q$ the cube $Q$ is one of the cubes in the supremum defining $Mf(x)$, so $Mf(x) \geq \frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy$, and therefore

$$\begin{aligned}
\left(\frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy\right)^p w(Q) &= \int_Q \left(\frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy\right)^p w(x)\,dx \\
&\leq \int_Q (Mf)^p\, w\,dx \\
&\leq \|Mf\|^p_{L^p_w} \\
&\leq C^p\,\|f\|^p_{L^p_w} \\
&= C^p \int_Q (w+\varepsilon)^{-\frac{p}{p-1}}\, w\,dx \\
&\leq C^p \int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dx.
\end{aligned}$$

Since $w$ is locally integrable, $w < \infty$ almost everywhere, so $(w+\varepsilon)^{-\frac{1}{p-1}} > 0$ almost everywhere and the integral on the right-hand side is positive. Dividing by it we obtain

$$\frac{w(Q)}{|Q|}\left(\frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy\right)^{p-1} \leq C^p.$$

As $\varepsilon \to 0$, $(w+\varepsilon)^{-\frac{1}{p-1}}$ increases to $w^{-\frac{1}{p-1}}$. Hence

$$\begin{aligned}
&\frac{w(Q)}{|Q|}\left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\,dy\right)^{p-1} \\
&\quad= \frac{w(Q)}{|Q|}\left(\frac{1}{|Q|}\int_Q \lim_{\varepsilon\to 0}(w+\varepsilon)^{-\frac{1}{p-1}}\,dy\right)^{p-1} \\
&\quad= \lim_{\varepsilon\to 0}\frac{w(Q)}{|Q|}\left(\frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy\right)^{p-1} \qquad \text{(monotone convergence)} \\
&\quad\leq C^p,
\end{aligned}$$

which is (2). Since $w(Q) > 0$, this also shows that $\int_Q w^{-\frac{1}{p-1}}\,dy < \infty$. □

# References


[1] Stephen M. Buckley, *Estimates for operator norms on weighted spaces and reverse Jensen inequalities*, Trans. Amer. Math. Soc. **340** (1993), no. 1, 253–272.

[2] Miguel de Guzmán, *Differentiation of integrals in* $\mathbb{R}^n$, Lecture Notes in Mathematics, vol. 481, Springer-Verlag, Berlin, 1975.

[3] Javier Duoandikoetxea, *Forty years of Muckenhoupt weights*, Function Spaces and Inequalities. Lecture Notes, Paseky nad Jizerou 2013 (Jaroslav Lukeš and Luboš Pick, eds.), Matfyzpress, Praha, 2013, pp. 23–75.

[4] Andrei K. Lerner, *An elementary approach to several results on the Hardy–Littlewood maximal operator*, Proc. Amer. Math. Soc. **136** (2008), no. 8, 2829–2833.

[5] Benjamin Muckenhoupt, *Weighted norm inequalities for the Hardy maximal function*, Trans. Amer. Math. Soc. **165** (1972), 207–226. MR 0293384